Ch 6.1: Subset Selection

Lecture 16 - CMSE 381

Prof. Elizabeth Munch

Michigan State University
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Dept of Computational Mathematics, Science & Engineering

Weds, Oct 9, 2024

Announcements

Last time

- k-fold CV for Classification

Covered in this lecture

- Subset selection
- Forward and Backward Selection

Announcements:

- HW #4 Due Tonight!

Lec #	Date			Reading	HW
12	Mon	9/30	Leave one out CV	5.1.1, 5.1.2	
13	Wed	10/2	k-fold CV	5.1.3	
14	Fri	10/4	More k-fold CV,	5.1.4-5	
15	Mon	10/7	k-fold CV for classification	5.1.5	
16	Wed	10/9	Subset selection	6.1	HW #4 Due Weds 10/9
17	Fri	10/11	Shrinkage: Ridge	6.2.1	
18	Mon	10/14	Shrinkage: Lasso	6.2.2	
19	Wed	10/16	Dimension Reduction	6.3	
20	Fri	10/18	Overflow, Possibly more dimension reduction?		HW #5 Due Fri 10/18
	Mon	10/21	No class - Fall break		
	Wed	10/23	**Review**		
	Fri	10/25	**Midterm #2**		

Section 1

Last time

Goals of fitting a given model

Up to now, we've focused on standard linear model: $Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \varepsilon$ and done least squares estimation.

Prediction accuracy ← want to do a good job at getting the right answer

— Let's talk about n vs p —

- If my true relationship is approx linear, then Least Sqs estimator have low bias
- If $n \gg p$, then least sqs estimates have low variance
- If n is not much larger than p, high variability, overfitting poor predictions (bad) (not many data pts)
- If $n < p$, then no unique solution, so can't use this anyway. Goal: Want to lower p!

Goals of fitting a given model

Up to now, we've focused on standard linear model: $Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \varepsilon$ and done least squares estimation.

Model Interpretability

↳ goal of understanding how the model is working

- Often many input variables not associated with the response
- Including them leads to unnecessary complexity

Goal today: get rid of unneeded variables

Section 2

Best Subset Selection

Too many variables

ex) $p=4$

All subsets of 4 variables ($2^4 = 16$)

- $\emptyset$
- X_1
- X_2
- X_3
- X_4
- $X_1\ X_2$
- $X_1\ X_3$
- $X_1\ X_4$
- $\boxed{X_2\ X_3}$
- $X_2\ X_4$
- $X_3\ X_4$
- $X_1\ X_2\ X_3$
- $X_1\ X_2\ X_4$
- $X_1\ X_3\ X_4$
- $X_2\ X_3\ X_4$
- $X_1\ X_2\ X_3\ X_4$

train $Y = \beta_0 + \beta_1 X_2 + \beta_2 X_3$
test on test data
k-fold CV $\Rightarrow$ test score

Naive idea:
Fit model for every option
pick the best option.

One way of breaking this up

Algorithm 6.1 *Best subset selection*

1. Let $\mathcal{M}_0$ denote the *null model*, which contains no predictors. This model simply predicts the sample mean for each observation.

   mean of y

2. For $k = 1, 2, \ldots p$:
   (a) Fit all $\binom{p}{k}$ models that contain exactly k predictors.
   (b) Pick the best among these $\binom{p}{k}$ models, and call it $\mathcal{M}_k$. Here *best* is defined as having the smallest RSS, ~~or equivalently largest R^2~~.

   training error

3. Select a single best model from among $\mathcal{M}_0, \ldots, \mathcal{M}_p$ using cross-validated prediction error, C_p (AIC), BIC, or adjusted R^2.

M_0
M_1
M_2
$\vdots$
M_p

K-fold CV (not the same k)

Calculate by hand

We train a model using four variables, X_1, X_2, X_3, X_4. We're interested in getting a subset of the variables to use. The following table shows the mean squared error and the MSE value computed for the model learned using each possible subset of variables.

		Training MSE (x10^7)	k-fold CV Testing Error
0	Null model	8.76	10.08
1	X1	8.63	9.98
	X2	7.42	8.01
	X3	8.16	8.3
	X4	8.33	9.06
2	X1,X2	4.33	7.47
	X1,X3	5.82	5.22
	X1,X4	3.17	4.23
	X2,X3	4.07	3.78
	X2,X4	3.31	4.01
	X3,X4	3.06	4.16
3	X1,X2,X3	3.08	5.49
	X1,X2,X4	3.55	4.02
	X1,X3,X4	2.97	4.23
	X2,X3,X4	2.98	3.17
4	X1,X2,X3,X4	2.16	4.39

$M_0 = \text{null}$

$M_1 = \{X_2\}$

$M_2 = \{X_3, X_4\}$

minimum of the M_k's

$M_3 = \{X_1, X_3, X_4\}$

$M_4 = \{X_1, X_2, X_3, X_4\}$

1. What subset of variables is found for each of the sets $\mathcal{M}_0, \mathcal{M}_1, \mathcal{M}_2, \mathcal{M}_3, \mathcal{M}_4$ when using best subset selection?
2. What subset of variables is returned using best subset selection?

$\hookrightarrow \{X_3, X_4\}$

Extra work space if it helps

	Training MSE (x10^7)	k-fold CV Testing Error
Null model	8.76	10.08
X1	8.63	9.98
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X1,X2,X3,X4	2.16	4.39

- $\emptyset$

- X_1
- X_2
- X_3
- X_4

- $X_1\ X_2$
- $X_1\ X_3$
- $X_1\ X_4$
- $X_2\ X_3$
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Code to do this

Section 3

Forward Selection

What's the problem?

Checking 2^P models is completely unreasonable

↳ Next goal:

Do this but more intelligently.

Forward Stepwise Selection

Algorithm 6.2 *Forward stepwise selection*

1. Let $\mathcal{M}_0$ denote the *null* model, which contains no predictors.
2. For $k = 0, \ldots, p-1$:
   (a) Consider all $p-k$ models that augment the predictors in $\mathcal{M}_k$ with one additional predictor.
   add one predictor from the X_i's
   (b) Choose the *best* among these $p-k$ models, and call it $\mathcal{M}_{k+1}$. Here *best* is defined as having smallest RSS ~~or highest R^2~~.
   training
3. Select a single best model from among $\mathcal{M}_0, \ldots, \mathcal{M}_p$ using cross-validated prediction error, C_p (~~AIC, BIC, or adjusted R^2~~.

test error

An example for Forward Stepwise Selection

- $\emptyset$

- X_1
- X_2
- X_3
- X_4

- $X_1\ X_2$
- $X_1\ X_3$
- $X_1\ X_4$
- $X_2\ X_3$
- $X_2\ X_4$
- $X_3\ X_4$

- $X_1\ X_2\ X_3$
- $X_1\ X_2\ X_4$
- $X_1\ X_3\ X_4$
- $X_2\ X_3\ X_4$

- $X_1\ X_2\ X_3\ X_4$

Group work: by hand same example with forward example

We train a model using four variables, X_1, X_2, X_3, X_4. We're interested in getting a subset of the variables to use. The following table shows the mean squared error and the R^2 value computed for the model learned using each possible subset of variables.

	Training MSE (x10^7)	k-fold CV Testing Error
Null model	8.76	10.08
X1	8.63	9.98
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X1,X2,X3,X4	2.16	4.39

1. What subset of variables is found for each of the sets $\mathcal{M}_0, \mathcal{M}_1, \mathcal{M}_2, \mathcal{M}_3, \mathcal{M}_4$ when using forward selection?
2. What subset of variables is returned using forward subset selection?

Extra work space if it helps

	Training MSE (x10^7)	k-fold CV Testing Error
Null model	8.76	10.08
X1	8.63	9.98
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X1,X2,X3,X4	2.16	4.39

$k=0$

$k=1$

options by adding one variable to M_1

null model

$\emptyset$

$M_0 = \phi$

X_1

X_2

X_3

X_4

$M_1 = \{X_2\}$

$X_1\ X_2$

$X_1\ X_3$

$X_1\ X_4$

$X_2\ X_3$

$X_2\ X_4$

$X_3\ X_4$

$M_2 = \{X_2, X_4\}$

$X_1\ X_2\ X_3$

$X_1\ X_2\ X_4$

$X_1\ X_3\ X_4$

$X_2\ X_3\ X_4$

$M_3 = \{X_2, X_3, X_4\}$

$X_1\ X_2\ X_3\ X_4$

$M_4 = \{X_1, X_2, X_3, X_4\}$

Best test error of M_i's is $\{X_2, X_3, X_4\}$

Pros and Cons of Forward Stepwise

Pros:

Computationally cheaper!

Number of models fit =

$$1 + \sum_{k=0}^{p-1}(p-k) = 1 + \frac{p(p+1)}{2}$$

$O(p^2)$ better than 2^p

Cons:

Not guaranteed to find best model

Not a con but limitation

↳ $n<p$

I can train upto # of data pts

$$\underbrace{M_0, M_1, M_2, \ldots, M_{n-1}}_{\text{can do through } n-1}\ [M_{n+1} \ldots M_p]$$

Section 4

Backward Selection

Backward stepwise selection

Algorithm 6.3 *Backward stepwise selection*

1. Let $\mathcal{M}_p$ denote the *full* model, which contains all p predictors.
2. For $k = p, p-1, \ldots, 1$:
   (a) Consider all k models that contain all but one of the predictors in $\mathcal{M}_k$, for a total of $k-1$ predictors.
   (b) Choose the *best* among these k models, and call it $\mathcal{M}_{k-1}$. Here *best* is defined as having smallest RSS ~~or highest R^2~~
   training error
3. Select a single best model from among $\mathcal{M}_0, \ldots, \mathcal{M}_p$ using cross-validated prediction error, ~~C_p (AIC), BIC, or adjusted R^2~~.

testing error

Pros and Cons of Backward Stepwise

Pros:

Comp. Cheaper

Same # models fit

$= O(p^2)$

$1 + p\frac{(p+1)}{2}$

Better than 2^p

Cons:

Not guaranteed best model

Vs forward selection:

$n < p$ $\left\{ \begin{array}{l} \text{Small \# of data pts} \\ \text{means can't start from} \\ M_p \text{ using all variables} \end{array} \right.$

↳ can't use it.

There are hybrid approaches combo of fwd and back...

TL;DR

Algorithm 6.1 *Best subset selection*

1. Let $\mathcal{M}_0$ denote the *null model*, which contains no predictors. This model simply predicts the sample mean for each observation.
2. For $k = 1, 2, \ldots p$:
   (a) Fit all $\binom{p}{k}$ models that contain exactly k predictors.
   (b) Pick the best among these $\binom{p}{k}$ models, and call it $\mathcal{M}_k$. Here *best* is defined as having the smallest RSS, or equivalently largest R^2.
3. Select a single best model from among $\mathcal{M}_0, \ldots, \mathcal{M}_p$ using cross-validated prediction error, C_p (AIC), BIC, or adjusted R^2.

- Modify step 2 with forward or backward selection
- Choose best model in step 3 using one of our adjusted training scores or CV

Next time

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